

Motion Revision

Multiple Choice Questions

KS3 Physics · 20 Questions · Arcadia British School Dubai

Instructions: For each question, circle the letter (A, B, C or D) that matches the correct answer. There is only one correct answer per question. Show any working in the space provided.

SPEED, DISTANCE & TIME

Question 1

What is the correct equation for speed?

A) Speed = distance $\times$ time

B) Speed = distance $\div$ time

C) Speed = time $\div$ distance

D) Speed = distance – time

Working / notes:

Question 2

A car travels 600 m in 30 seconds. What is its speed?

A) 0.05 m/s

B) 18 000 m/s

C) 20 m/s

D) 570 m/s

Working / notes:

Question 3

Sophie runs at 8 m/s for 25 seconds. How far does she travel?

A) 3.125 m

B) 17 m

C) 200 m

D) 33 m

Working / notes:

Question 4

How long does it take an object to travel 450 m at a speed of 15 m/s?

A) 30 s

B) 6750 s

C) 0.03 s

D) 435 s

Working / notes:

Question 5

Which of the following is the approximate walking speed of a person?

- A) 0.15 m/s
- B) 1.5 m/s
- C) 15 m/s
- D) 150 m/s

Working / notes:

Question 6

Eliza walks 2000 m in 50 minutes. What is her speed in m/s?

- A) 40 m/s
- B) 0.67 m/s
- C) 2.5 m/s
- D) 6.7 m/s

Working / notes:

DISTANCE-TIME GRAPHS

Question 7

What does a horizontal (flat) line on a distance-time graph represent?

- A) The object is accelerating
- B) The object is moving at constant speed
- C) The object is stationary
- D) The object is decelerating

Working / notes:

Question 8

On a distance-time graph, what does the gradient (slope) of the line tell you?

- A) Acceleration
- B) Speed
- C) Distance
- D) Time

Working / notes:

Question 9

A line on a distance-time graph goes from 0 m at $t = 0$ s to 30 m at $t = 10$ s. What is the speed?

- A) 0.33 m/s
- B) 3 m/s
- C) 30 m/s
- D) 300 m/s

Working / notes:

Question 10

A steeper gradient on a distance-time graph indicates...

- A) A lower speed
- B) The object is stationary
- C) A greater speed
- D) The object is decelerating

Working / notes:

Question 11

A distance-time graph shows the distance decreasing back towards zero. What is happening?

- A) The object is speeding up
- B) The object is returning towards the start
- C) The object is stationary
- D) The object has a positive velocity

Working / notes:

ACCELERATION

Question 12

What is the correct equation for acceleration?

- A) $a = v \times t$
- B) $a = (v + u) \div t$
- C) $a = (v - u) \div t$
- D) $a = v \div u$

Working / notes:

Question 13

What is the unit for acceleration?

- A) m/s
- B) m/s^2
- C) m
- D) s

Working / notes:

Question 14

A car speeds up from 5 m/s to 25 m/s in 10 seconds. What is its acceleration?

- A) 2.5 m/s^2
- B) 30 m/s^2
- C) 2 m/s^2
- D) 0.5 m/s^2

Working / notes:

Question 15

A ball decelerates from 20 m/s to 0 m/s in 5 seconds. What is the magnitude of its deceleration?

- A) 100 m/s^2
- B) 4 m/s^2
- C) 25 m/s^2
- D) 0.25 m/s^2

Working / notes:

Question 16

A rocket accelerates at 500 m/s^2 for 6 seconds from rest. What is its final velocity?

- A) 83.3 m/s
- B) 506 m/s
- C) 3000 m/s
- D) 494 m/s

Working / notes:

Question 17

Which of the following best describes deceleration?

- A) An increase in speed
- B) Negative acceleration — the object is slowing down
- C) Moving at a constant velocity
- D) A change in direction only

Working / notes:

VELOCITY-TIME GRAPHS

Question 18

What does an upward-sloping line on a velocity-time graph represent?

- A) Constant speed
- B) Deceleration
- C) The object is stationary
- D) Acceleration (speeding up)

Working / notes:

Question 19

What does the area under a velocity-time graph represent?

- A) Acceleration
- B) Speed
- C) Distance travelled
- D) Time

Working / notes:

Question 20

A velocity-time graph shows a straight line from 0 m/s at $t = 0$ s to 60 m/s at $t = 20$ s. What is the acceleration?

A) 1200 m/s^2

B) 3 m/s^2

C) 0.33 m/s^2

D) 80 m/s^2

Working / notes:

Question 21

An object travels at 40 m/s for 10 seconds (flat line on a V-T graph). How far does it travel?

A) 4 m

B) 50 m

C) 400 m

D) 0.25 m

Working / notes:

ANSWER KEY

Motion Revision — 20 Multiple Choice Questions

Q	Ans	Topic	Q	Ans	Topic
1	B	Speed, Distance & Time	11	B	Distance-Time Graphs
2	C	Speed, Distance & Time	12	C	Acceleration
3	C	Speed, Distance & Time	13	B	Acceleration
4	A	Speed, Distance & Time	14	C	Acceleration
5	B	Speed, Distance & Time	15	B	Acceleration
6	B	Speed, Distance & Time	16	C	Acceleration
7	C	Distance-Time Graphs	17	B	Acceleration
8	B	Distance-Time Graphs	18	D	Velocity-Time Graphs
9	B	Distance-Time Graphs	19	C	Velocity-Time Graphs
10	C	Distance-Time Graphs	20	B	Velocity-Time Graphs

EXPLANATIONS

SPEED, DISTANCE & TIME

Q1 Answer: B — Speed = distance ÷ time

Speed = distance ÷ time ($v = s \div t$). Distance is on top of the formula triangle.

Q2 Answer: C — 20 m/s

$v = s \div t = 600 \div 30 = 20$ m/s

Q3 Answer: C — 200 m

$s = v \times t = 8 \times 25 = 200$ m

Q4 Answer: A — 30 s

$t = s \div v = 450 \div 15 = 30$ s

Q5 Answer: B — 1.5 m/s

Typical walking speed ≈ 1.5 m/s (about 5 km/h).

Q6 Answer: B — 0.67 m/s

Convert time: 50 min = 3000 s. $v = 2000 \div 3000 = 0.67$ m/s

DISTANCE-TIME GRAPHS

Q7 Answer: C — The object is stationary

A flat line means distance is not changing — the object is not moving (stationary).

Q8 Answer: B — Speed

Gradient = rise ÷ run = change in distance ÷ change in time = speed.

Q9 **Answer: B — 3 m/s**
Speed = gradient = $30 \div 10 = 3 \text{ m/s}$

Q10 **Answer: C — A greater speed**
Steeper gradient = greater rise $\div$ run ratio = greater speed.

Q11 **Answer: B — The object is returning towards the start**
A downward slope means distance from the start is decreasing — the object is heading back.

ACCELERATION

Q12 **Answer: C — $a = (v - u) \div t$**
Acceleration = change in velocity $\div$ time = $(v - u) \div t$, where v = final velocity and u = initial velocity.

Q13 **Answer: B — m/s^2**
Acceleration is measured in metres per second squared (m/s^2) because it is a change in velocity per unit of time.

Q14 **Answer: C — 2 m/s^2**
 $a = (v - u) \div t = (25 - 5) \div 10 = 20 \div 10 = 2 \text{ m/s}^2$

Q15 **Answer: B — 4 m/s^2**
 $a = (0 - 20) \div 5 = -4 \text{ m/s}^2$. Magnitude of deceleration = 4 m/s^2

Q16 **Answer: C — 3000 m/s**
 $v = u + at = 0 + 500 \times 6 = 3000 \text{ m/s}$

Q17 **Answer: B — Negative acceleration — the object is slowing down**
Deceleration is negative acceleration — velocity is decreasing over time.

VELOCITY-TIME GRAPHS

Q18 **Answer: D — Acceleration (speeding up)**
An upward slope means velocity is increasing with time — the object is accelerating.

Q19 **Answer: C — Distance travelled**
Area under a velocity-time graph = distance travelled. Break the area into shapes (triangles, rectangles) and sum them.

Q20 **Answer: B — 3 m/s^2**
Acceleration = gradient = $(60 - 0) \div (20 - 0) = 60 \div 20 = 3 \text{ m/s}^2$

Q21 **Answer: C — 400 m**
Area = rectangle = base $\times$ height = $10 \times 40 = 400 \text{ m}$
